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MILITARY SPECIFICATION

MAGNESIUM OXIDE, CALCINED (FOR USE IN AMMUNITION)

Inactive for new design after 06May14, and is no longer used except for replacement purposes.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of three types of calcined magnesium oxide (see 6.1).

Type I – for use in aerial mines
Type II – for use in M864 propellant
Type III – for use in XM1128 propellant

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC – AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 300 - Standard Practice for Sampling Industrial Chemicals.

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Physical and chemical requirements. The material shall be magnesium oxide which conforms to the following requirements.

TABLE I. Chemical and physical requirements for magnesium oxide.

Property	Type I	Type II	Type III
Iodine number, meq/100g, minimum (min.)	85	----	----
Iron, as iron oxide (Fe_2O_3), %, maximum (max.)	0.18	----	----
Loss on ignition, %, max.	10.0	10.0	10.0
Calcium, as calcium oxide (CaO), %, max.	1.00	1.0	1.00
Magnesium, as magnesium oxide (MgO), %, min.	97.2	88.0	88.0
Silica, as silica dioxide (SiO_2), %, max.	0.34	----	----
Bulk density, loose, lb/ft^3, min.	8.0	----	----
Granulation Thru US Alternate sieve number 325, wet, min. Thru US Alternate sieve number 200, wet, min.	 99.5% 99.8%	 99.9% ----	 99.9% ----
Aluminum, as aluminum oxide (Al_2O_3), %, max.	0.15	----	----
Moisture, %, max.	----	0.5	0.5
Surface area (BET) (m^2/g), min.	----	----	145.0

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3.2 Workmanship. The magnesium oxide shall be a fluffy, white powder and shall be free of lumps and foreign matter.

3.3 First article. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

4. VERIFICATION

4.1 DELETED.

4.1.1 DELETED.

4.1.2 DELETED.

4.1.3 First article inspection.

4.1.3.1 Submission. The contractor shall submit a first article quantity as designated by the contracting officer for evaluation in accordance with provisions of 4.1.3.2 (see 6.2). The first article sample shall consist of 500 grams of the material selected randomly in a manner similar to the method described in 4.2.2. The sample submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.1.3.2 Inspections to performed. The sample will be subjected by the Government to any or all of the examinations or specified in 4.2 and 4.3.

4.1.3.3 Rejection. If the sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.2 Inspection provisions.

4.2.1 Lot formation. A lot of magnesium oxide shall consist that quantity produced, by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of magnesium oxide that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. In the event the process is a batch operation, each batch shall constitute a lot. Lot formation shall be in accordance with MIL-STD-1916, Formation and identification of lots or batches.

4.2.2 Sampling.

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4.2.2.1 Primary samples. Ten percent or a minimum of 25, whichever is less, of the containers comprising each lot submitted shall be selected at random. A representative one-pint sample from each selected container shall be taken in the following manner: insert a closed sample tube to maximum depth, at an oblique angle, then open and agitate to fill the tube. Each sample shall be placed in a clean, dry, glass jar, which shall be labeled to identify the container, lot number, contract number, and material name.

4.2.2.2 Composite samples. The primary sample shall be divided into two groups of equal size, and a composite sample shall be prepared from each group. Each composite sample shall be placed in a clean, dry container, sealed, and labeled in a manner that will identify the sample with the contract number, material name, containers, and lot represented. Each composite sample shall be subjected to all chemical and physical tests specified. If either sample fails to meet the requirements specified in 3.1 and 3.2, the lot shall be rejected.

4.2.2.3 Sampling for M864 propellant component. Approximately 500 grams of the composition shall be selected from each batch to be samples using ASTM E 300 for solids. Samples shall be selected for inspection in accordance with table below. If any sample fails to meet any test requirement the batch represented shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used. All characteristics are defined as Major in accordance with MIL-STD-1916.

TABLE II. Sampling for M864 propellant component.

Number of containers in batch	Number of containers to be sampled
2 – 200	2
Over 200	5

4.3 Test methods and procedures. All tests given in this section shall be performed using prescribed procedures for replicate determination given in standard analytical chemistry textbooks. Also, unless otherwise specified herein, all chemicals and reagents used in the tests shall be ACS Grade or Reagent Grade. Purified and distilled water in accordance with ASTM D 1193 shall be used as required in all test procedures.

4.3.1 Iodine number (Type I). Determine iodine number of Type I material in accordance with the following procedure.

4.3.1.1 Reagents.

a. Iodine solution in aliphatic naphtha, 0.10N
b. Standard sodium thiosulfate solution, 0.10N
c. Potassium iodide solution in 75% ethanol, 0.03N

4.3.1.2 Apparatus.

a. Automatic pipette, 100-mL, gravity-fed from iodine solution reservoir equipped with a drying tube on the air inlet
b. Burrell “wrist-action” shaker, or equivalent
c. Pipette, 20-mL, stopcock-type.

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To maintain adequacy and precision in this method, which is dependent on the measurement of the differences between two relatively large quantities, follow closely the prescribed details of the technique, which are designed to minimize the introduction of moisture from the atmosphere and from absorbed films on glassware, and to avoid uncompensated absorption of iodine on glass surfaces. Glassware must be cleaned with acid to remove minute deposits of basic magnesium compounds.

4.3.1.3 Procedure. Weigh rapidly a 2.0 gram sample and transfer the sample quantitatively, through a powder funnel, into a 100-mL cylinder. Rinse the automatic pipette with 0.10N iodine solution and discard the rinse. Immediately refill the pipette, and add 100 mL of the iodine solution to the sample. Prepare a blank to be run in parallel with the sample. The rinse need not be repeated if the solution is added to subsequent samples without delay. Secure the glass stopper in each cylinder with a clip, and center the cylinder horizontally in the mounting clamp on the shaker. Shake the cylinder at maximum rate for thirty minutes, remove from the shaker, and allow contents to settle for ten minutes. From the top third of the clear supernatant liquid in each cylinder, transfer a 20 mL aliquot, using a clean, dry stopcock pipette (the pipette may be air dried with a rubber bulb between determinations) to a 250-mL Erlenmeyer flask or equivalent, containing 50 mL of 0.03N alcoholic potassium iodide solution and titrate with standard 0.10N sodium thiosulfate. Calculate the iodine number as follows:

$$\text{Iodine number} = \frac{5 \text{ x } (A - B) \text{x } N}{W} \text{ x } 100$$

Where: A = sodium thiosulfate solution used for the blank, mL
B = sodium thiosulfate solution used for the sample, mL
N = normality of the sodium thiosulfate solution
W = weigh of sample, g

4.3.2 Iron, as iron oxide (Fe_2O_3). Determine iron oxide content in accordance with either of the following methods.

4.3.2.1 Spectrophotometer method (preferred).

4.3.2.1.1 Reagents.

a. H2SO4 solution, 50%
b. Hydroxlamine hydrochloride, 10%, freshly made
c. Sodium citrate ($Na_2C_6H_5O_7 \bullet 2\ H_2O$), 25% solution
d. HCl, concentrated
e. Nitric acid (HNO_3), concentrated
f. Bright iron wire, primary standard grade

4.3.2.1.2 Apparatus.

a. Beckman spectrophotometer, Model DU, or equivalent, with 3 matched 1-cm corex cells, set up for use with tungsten lamp and red sensitive phototube, without filter.

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4.3.2.1.3 Standard preparation. Weigh accurately 0.05 g of bright iron wire (an equivalent weight of any other recognized primary standard, with appropriately modified preparation procedure, may be substituted for the iron wire standard) and place it in a 1-L volumetric flask, add 2 mL of concentrated HCl and a few mL of hot distilled water. When all the iron is dissolved, fill the flask to volume, and mix. Transfer 1.00, 3.00, 5.00, 7.00, and 9.00 mL portions of the standard iron solution to a series of 100-mL volumetric flasks, and prepare a blank. Dilute each to approximately 70 mL and determine the absorbency (A_s) by the procedure described below for analysis of the sample. The initial calibration should consist of four complete determinations, two on each of two days, preferably one by each of two analysts. The calibration should be checked by two additional single determinations for each new bottle of solid O-phenanthroline reagent. Calculate the concentrations, plot them against the measured absorbances, and draw the line of best fit. A line that deviates significantly from the zero intercept or shows curvature indicates error. Calculate numerically the absorptivity as follows:

$$a_s = \frac{\sum(a_s \times b \times C) - (\sum A_s) \times \frac{\sum b \times C}{n}}{\sum(b \times C)^2 - \frac{\sum b \times C^2}{n}}$$

Where: a_s = absorptivity, mL/(g-cm) (2.03×10^5 mL/(gm-cm) for given apparatus
A_s = absorbance
C = concentration off iron in diluted aliquot, $\frac{g}{mL} = \frac{W \times V}{100{,}000}$
W = weight of primary standard, g
V = volume of standard solution used, mL
b = length of light path through the cell, cm
n = number of determinations

4.3.2.1.4 Sample procedure. Weigh 1 g of sample and transfer it to a 50-mL beaker. Add 20 mL of distilled water, dissolve the sample by dropwise addition of concentrated HCl until the solution is just acid to litmus, then add 1 mL of acid in excess, and prepare a blank to run parallel to the sample. Bring to a gentle boil to insure complete solution, then cool. Transfer the solution to a 100-mL volumetric flask, and dilute to approximately 70 mL with distilled water. Treat both sample and blank as follows: Add 1 mL of freshly prepared hydroxylamine hydrochloride solution, mix well, and let the mixture stand for at least 15 minutes. Add 10 mL of 0.25% o-phenanthroline solution and mix well. Add 1 mL of 25% sodium citrate solution, dilute to volume, mix well, and allow to stand for at least 30 minutes at not less than 20˚C. Using the blank for comparison, determine the absorbance of the sample solution with spectrophotometer at a wavelength of 510 millimicrons and a slit width of 0.18 mm. Calculate the percent of Fe_2O_3 as follows:

$$Fe_2O_3, \% = \frac{79.8 \times 10^6 \times A_s}{55.8 \times a_s \times b \times w \times v} = \frac{14.3 \times (10^5) \times A_s}{A_s \times b \times w \times v}$$

Where: a_s = absorptivity, mL/(g-cm)
A_s = absorbance of sample solution
b = cell length, cm

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v = volume of aliquot, mL
w= sample weight, g

4.3.2.2 Alternate spectrophotometer method.

4.3.2.2.1 Apparatus and reagents.

a. Beckman spectrophotometer, model DU, or equivalent
b. pH meter
c. Ferron reagent, 0.2% solution. Dissolve 0.40 g reagent grade ferron (8-hydroxy-7-iodo-5-quinolinesulfonic acid) in deionized water and dilute to 200 mL (complete dissolution is obtained by heating to 80˚C).
d. Ammonium acetate buffer, 10% solution. Dissolve 50.0 g of reagent grade ammonium acetate in deionized water and dilute to 500 mL.
e. Nitric acid, 1:9 solution. Add 10 mL concentrated nitric acid to 90 mL deionized water.
f. Magnesia solution, 1 g MgO/100 mL concentrated nitric acid to 90 mL deionized water.
g. Stock iron standard, 20 µg Fe_2O_3/mL. Dissolve 0.0140 g of electrolytic iron wire in 10 mL concentrated hydrochloric acid and 5 mL nitric acid and evaporate almost to dryness. Dissolve in deionized water and dilute to 1 liter.
h. Stock aluminum standard, 20 µg Al_2O_3/mL. Dissolve 0.106 g reagent grade aluminum turnings in 10 mL concentrated hydrochloric acid and evaporate almost to dryness. Dissolve in deionized water, dilute to 1 liter in a volumetric flask and mix thoroughly.
i. Hot plate, gas two burners
j. Buffer solution, pH 4.00
k. Buffer solution, pH 10.00
l. Pipets, volumetric: 10, 20, 25, 30-mL
m. Burets, Schellbach automatic zero

4.3.2.2.2 Preparation of standards.

1. Into four 150-mL Griffin beakers introduce, by pipet, 5, 15, 25, and 50 mL of the aluminum stock solution (100, 300, 500, and 1000 µg Al_2O_3).

Into four 150-mL Griffin beakers introduce 5, 15, 25, and 50 mL of the iron stock solution (100, 300, 500, and 1000 µg Fe_2O_3).

Use and empty 150-mL beaker for a reagent blank. This can be used for both standard series. Mark each beaker distinctly to identify its contents.

2. To each of the beakers, add the following reagents in the order and quantities given:

25 mL magnesia solution
4 mL 1:9 HNO_3
Sufficient 10% ammonium acetate solution to bring the pH to 5.2 (this will require about 25 mL) using a pH meter.

3. Transfer the contents of each beaker to a 200-mL volumetric flask, rinsing the beaker thoroughly and adding the rinse water to the volumetric flask. Add 10 mL of

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ferron reagent, dilute to the mark with deionized water and mix thoroughly. Allow the solutions to stand for 15 minutes for color development.

4. Place each of the standards and the blank in a 1.00 cm absorption cell, and using the blank as reference, measure the absorbance of each iron solution at 370 µm and 600 µm; and each aluminum solution at 370 µm, with the Beckman DU spectrophotometer and prepare standard curves as described above.

4.3.2.2.3 Sample procedure.

a. Weigh a 1.000 g sample and transfer to a 250-mL beaker, cover with 5 mL of deionized water, and add 6-10 mL of concentrated hydrochloric acid. Warm on a hot plate at low heat until the sample has dissolved. Transfer the solution to a 100-mL volumetric flask, rinsing the beaker thoroughly. Dilute to the mark.

b. Transfer a 25 mL aliquot of the above solution by pipet to a 150-mL beaker and 4 mL of 1:9 nitric acid to oxidize all iron. To a second 150-mL beaker, add 25 mL of standard MgO solution and carry through all subsequent steps as a reagent blank.

c. Place the electrodes of a pH meter into the solution and, by means of a 50-mL buret, add 10% ammonium acetate buffer until a pH of exactly 5.2 is obtained. Rinse the electrodes into the beaker with deionized water. Rinse the solution into a 200-mL volumetric flask with deionized water.

d. Add 10 mL of ferron reagent by pipet and dilute to the mark with deionized water. Invert the volumetric flask repeatedly to mix. Allow to stand for 15 minutes for color development.

e. Fill a 1.000 Beckman cell with the sample solution, and a second with the blank. Place the cells in the carrier of the Beckman DU spectrophotometer.

f. Put the tungsten lamp in place, and make suitable settings for measuring at 370 µm as described in Table III.

TABLE III. Instrument settings.

Settings	Type of instrument			
	With photomultiplier		Without photomultiplier	
Wavelength, µm	370	600	370	600
Slit width, mm	0.2	0.025	0.2	0.05
Photo tube	P-M	P-M	Blue	Blue
Filter	Corex	None	Corex	None
Resistor, megohms	22	22	2000	2000
Check switch	0.1	0.1	0.1	0.1
P-M sensitivity	3	3	-	-

g. Turn the tungsten lamp on and the selector switch to “check.” With the shutter switch closed, zero the galvanometer with the dark current control.

h. Place the blank solution in the light path, open the shutter switch and again zero the galvanometer, this time using the sensitivity control.

i. Turn the selector switch to 0.1 and place the sample solution in the light path. Balance the galvanometer with the transmittance control. Read the absorbance from the scale, and record as “absorbance at 370 µm.”

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j. Make the necessary settings for measuring at 600 µm. Repeats steps g through i, and record this reading as "absorbance at 600 µm."

k. From the standard curve for aluminum at 370 µm, using the "absorbance at 370 µm," obtain and record "total uncorrected aluminum."

l. Using the iron concentration obtained in step 1 obtain the "iron absorbance at 600 µm."

m. Using the iron concentration obtained in step l obtain the "iron absorbance at 370 µm" from the iron standard curve at 370 µm.

n. Using the iron absorbance at 370 µm, obtain from the aluminum standard curve, the equivalent aluminum concentration. Subtract this concentration from the "total uncorrected aluminum," step k. Record as "true aluminum."

4.3.2.2.4 Calculations. Since both the standard and the sample are diluted to the same volume, the results are directly comparable. Thus, the amount of Fe_2O_3 and Al_2O_3 are given by the following equations:

$$Fe_2O_3, \% = \frac{\mu g\ Fe_2O_3\ found}{g\ sample} \times \frac{1\ g}{1{,}000{,}000\ \mu g} \times \frac{100\ mL\ dilution}{25\ mL\ aliquot} \times 100$$

$$Al_2O_3, \% = \frac{\mu g\ Al_2O_3\ (true\ Al)}{g\ sample} \times \frac{1\ g}{1{,}000{,}000\ \mu g} \times \frac{100\ mL\ dilution}{25\ mL\ aliquot} \times 100$$

When taking a 1.000 g sample, diluting to 100 mL and taking a 25 mL aliquot the calculations become:

$$Fe_2O_3, \% = \mu g\ Fe_2O_3\ found \times 0.004$$

$$Al_2O_3, \% = \mu g\ Al_2O_3\ (true\ Al) \times 0.004$$

4.3.3 Loss on ignition. Determine the loss on ignition in accordance with the following.

4.3.3.1 Apparatus.

a. Muffle furnace
b. Analytical balance
c. Two porcelain crucibles, Coors No. 1 (tall form)
d. Desiccator

4.3.3.2 Procedure. Ignite a crucible for one hour in the muffle furnace at 1600˚F. Remove the crucible and allow to cool to room temperature in a desiccator. Weigh the crucible and record the weight to the nearest 0.0001 g. Place approximately one (1) g of sample in the crucible and weigh to the nearest 0.0001 g. Ignite in the muffle furnace at 1600˚F for one hour. Remove and cool to room temperature in a desiccator. Weigh and record the weight. Calculate the loss on ignition as follows:

$$\text{Loss on ignition}, \% = \frac{\text{Weight loss, g}}{\text{Sample weight, g}} \times 100$$

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NOTE: Prepare an additional amount of ignited sample for use in 4.4.5.2 Calcium as Cao and 4.3.5 Magnesium as MgO.

4.3.4 Calcium, as calcium oxide (CaO). Determine the CaO content in accordance with the following methods, as applicable.

4.3.4.1 Type I (preferred method) and Type III.

4.3.4.1.1 Reagents.

a. Sulfuric acid, 25%, reagent grade
b. Ethanol, reagent grade

4.3.4.1.2 Procedure. Except for 0.5 g (for use in 4.3.5 Magnesium as MgO), transfer the remainder of the ignited sample obtained in 4.3.3 Loss on ignition to a tared 30-mL beaker and weigh. Add 4 mL of water, then stir and add HCl dropwise until the sample is just dissolved. Dilute to 10 mL with water and filter through a Whatman #40 filter paper, or equivalent, into a 250-mL beaker. Add to the filtrate 100 mL of ethanol and 40 mL of 25% H_2SO_4. Stir well and allow to stand over night. Filter the liquid through an ignited (400 to 500˚C) and tared Coors crucible, or equivalent. Transfer and wash the precipitate with a mixture of 2 volumes of ethanol to 1 volume of 25% H_2SO_4. Dry the crucible and precipitate at 105˚C for 30 minutes with the crucible supported on a ceramic plate (to prevent reaction of the retained acid with the metal shelf of the oven), then ignite at 400 to 500˚C for thirty minutes. Cool in a desiccator and weight. Calculate the percent of CaO as follows:

$$\text{CaO, percent} = \frac{56.08 \text{ x A}}{136.14 \text{ x W}} = \frac{0.412 \text{ x A}}{\text{W}}$$

Where: A = weight of calcium sulfate, g
W = weigh of ignited sample, g

4.3.4.2 Type I (alternate method).

4.3.4.2.1 Apparatus and reagents.

a. Analytical balance
b. Crucibles, Gooch, porcelain, No. 3
c. Asbestos fiber, analytical grade
d. Hydrochloric acid, concentrated, reagent grade
e. Ammonium hydroxide, concentrated, reagent grade
f. 6N sulfuric acid (1:6)
g. 0.1N potassium permanganate, standardized
h. 4% ammonium oxalate solution
i. 5% HCl-NH_4Cl solution – 50 g ammonium chloride plus 50 mL concentrated hydrochloric acid diluted to 1 liter with deionized water
j. Methanol, absolute, reagent grade
k. Methyl red indicator

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4.3.4.2.2 Procedure. Weigh approximately 3 g of sample on the analytical balance to the nearest milligram and transfer to a 250-mL beaker. Add 10 mL deionized water and 20 mL concentrated HCl, with swirling, and evaporate the sample to dryness. Cool the beaker and then add 5 mL 6N sulfuric acid. Cover the beaker with a watch glass and heat until all salts are dissolved. Cool slightly and add 50 mL absolute methanol with thorough mixing. Filter the solution through a prepared Gooch crucible with an asbestos mat, washing the beaker with five 10.0 mL portions of methanol. Transfer the asbestos mat with the precipitate to the original beaker, washing with 35 mL of 5% HCl-NH_4Cl solution. Continue washing the crucible with deionized water until the volume in the beaker is about 100 mL. Add 3 drops of methyl red indicator and heat to boiling. While still hot add 25 mL 4% ammonium oxalate solution with stirring and then add concentrated ammonium hydroxide dropwise until the solution turns yellow. Allow to stand in a warm place for one hour. Filter through another prepared Gooch crucible with an asbestos mat, washing the beaker with five 10.0 mL portions of 0.1% ammonium oxalate solution. Transfer entire crucible with the precipitate to a 250-mL beaker, add 150 mL 10% sulfuric acid, and heat almost to boiling. Titrate while hot with 0.1N potassium permanganate to a permanent pink endpoint. Calculate the calcium oxide concentration as follows:

$$\%CaO = \frac{\text{mL } KMnO_4 \text{ x N of } KMnO_4 \text{ x } 0.02804}{\text{sample weight}} \text{ x } 100$$

4.3.4.3 Calcium oxide (Type II). Dissolve 0.630 g of magnesium oxide in a mixture of 3 mL of concentrated H_2SO_4 and 22 mL of water. Add 50 mL of ethyl alcohol and allow to stand for a minimum of 16 hours. Warm to 50˚C and transfer the crystals of calcium sulfate to a Gooch crucible with an asbestos mat that has been washed with a 10% H_2SO_4 solution, water and ethyl alcohol and has been ignited to constant weight at 800˚C. Wash the crystals in the filter with a mixture of 2 volumes of ethyl alcohol and 1 volume of 10% H_2SO_4. Ignite the crucible and contents for 30 minutes at 800˚C. Cool and weigh. Calculate the percent calcium oxide as follows:

$$\text{Weight CaO} = \text{Weight } CaSO_4 \text{ x } 0.412$$

$$\%\ \text{CaO} = \frac{\text{weight CaO determined}}{\text{weight of sample}} \text{ x } 100$$

4.3.5 Magnesium, as magnesium oxide (MgO). Determine the MgO content in accordance with the following.

4.3.5.1 Reagents.

a. Standard HCl solution, 1.0N
b. Standard NaOH solution, 1.0N
c. Methyl orange indicator

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4.3.5.2 Procedure. Weigh accurately the appropriate 0.5 g of ignited sample (see 4.3.4 Calcium as CaO) into a 250-mL Erlenmeyer flask or equivalent. Dissolve the sample in 50.0 mL of standard 1.0N HCl. Back titrate the excess acid with standard 1.0N NaOH, using methyl orange indicator. Calculate the percent of MgO as follows:

$$\text{MgO, \%} = (100 \text{ x } \frac{((\text{A x } N_a) - (\text{B x } N_b))}{1{,}000\text{xW}}) - (40.32 \text{ x } \left(\frac{\text{C}}{56.08} + \frac{\text{D}}{53.0}\right)$$

$$= 2.016 \text{ x } (\frac{\text{A x } N_a - \text{ B x } N_b}{\text{W}}) - (0.72 \text{ x C}) - (0.76 \text{ x D})$$

Where: A = volume of standard HCl, mL
B = volume of standard NaOH, mL
C = CaO, % (from 4.4.1.9.2)
D = Na_2CO_3, % (see 4.5.1.3 below)
N_a = normality of standard HCl
N_b = normality of standard NaOH
W = weight of ignited sample

4.3.5.3 Sodium bicarbonate correction. The determination of Na_2CO_3 shall be made as follows: Weigh 5.0 g of sample, add 75 mL of water, and boil for 5 minutes. Filter with mild vacuum through an 80 mm funnel (Buchner or equivalent), fitted with Whatman No. 2 filter paper. Compare the filtrate with distilled water, and if any cloudiness is noted, filter again through a Coors No. 3001 filtering crucible or equivalent. Cool the filtrate, transfer to a 100-mL volumetric flask, dilute to mark, mix, and titrate a 20-mL aliquot with 0.10N HCl using methyl red as indicator. Calculate the percent Na_2CO_3 as follows:

$$Na_2CO_3 = \frac{5 \text{ x A x N x } 53.0 \text{ x } 100}{\text{weight of sample x } 1000} = \frac{\text{A x N x } 26.5}{\text{weight of sample}}$$

4.3.6 Silica, as silica dioxide (SiO_2). Determine silica dioxide content in accordance with the following method.

4.3.6.1 Apparatus and reagents.

a. Muffle furnace (1600˚F)
b. Analytical balance
c. Hot plate
d. Laboratory oven (110˚C)
e. Desiccator
f. Porcelain crucibles, Coors No. 1 (tall form)
g. Concentrated hydrochloric acid, reagent grade
h. Acetic anhydride, reagent grade
i. Filter paper, Whatman No. 30, ashless

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4.3.6.2 Procedure. Weigh approximately 1 gram of sample to the nearest 0.001 g. Transfer the material to a 250-mL beaker and add 5 mL deionized water and 20 mL concentrated hydrochloric acid. Place beaker on a hot plate, break up all gritty particles with a stirring rod, and cover with a watch glass. Evaporate the sample to dryness and add 10 mL acetic anhydride and 10 mL deionized water to the slightly cooled residue. Heat on the hot plate. A mild reaction indicates sufficient heating. At this point place the sample in a warm place until the reaction ceases. Add 4 mL of 1:1 hydrochloric acid and 20 mL of deionized water. Mix thoroughly. Filter the sample through Whatman No. 30 ashless filter paper. Wash the residue with 25 mL of hot acidified water (5 mL concentrated hydrochloric acid to 1 liter deionized water) and then wash thoroughly with hot deionized water. Transfer the filter paper to a previously ignited, tared crucible. Char the filter paper with a blast lamp and ignite for one hour in the muffle furnace at 1600˚F. Cool and reweigh the crucible. The weight difference is the weight of the precipitate. Calculate the silica concentration as follows:

$$\%SiO_2 = \frac{\text{weight of precipitate} - \text{weight of filter ash}}{\text{weight of initial sample}} \text{ x } 100$$

4.3.7 Bulk density, loose. Determine bulk density in accordance with the following procedure.

4.3.7.1 Apparatus.

a. Laboratory screen, U.S. Alternative No. 20 in accordance with ASTM E 11

4.3.7.2 Procedure. Weigh a 100 mL cylinder (a 250-ml cylinder which has been cut off at the 100-mL mark) to the nearest 0.01 g. Position the powder funnel above the cylinder so that its bottom tip is on inch above the cylinder rim. Screen the sample through a U.S. Alternative No. 20 sieve to avoid any lumps. Slowly shake the powder from a spatula so that it falls onto the side walls of the funnel. Avoid any vibration. Fill the cylinder to heaping. Carefully scrape the cylinder rim with the spatula so as to level the contents in a single stroke. Carefully remove any material clinging to the outside of the cylinder. Reweigh the filled cylinder and calculate the bulk density as follows:

$$\text{Bulk density } \left(\frac{\text{lb}}{\text{ft}^3}\right) = \frac{\text{weigh of full cylinder} - \text{weight of empty cylinder}}{\text{cylinder volume}} \text{ x } 62.4$$

4.3.8 Granulation. Determine granulation in accordance with the following procedure.

4.3.8.1 Apparatus and reagents.

a. U.S. Alternative screen No. 325 with pan
b. U.S. Alternative screen No. 200 with pan
c. Triple beam balance
d. Analytical balance
e. Paint brush, ½ inch wide by 5/8 inch long, with pure bristles
f. Laboratory oven
g. Graduated cylinder, 500 mL
h. Separator funnel, 1 1 (globe-shaped)

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i. Tubing, Tygon
j. Acetone

4.3.8.2 Wet screen test. Weigh 10±0.1 g of sample on a triple-beam balance and transfer to the U.S. Alternative No. 325 (and No. 200) screen. Clamp the screen under the globe-shaped funnel which has a tygon tube (1/2 foot to 1 foot in length and ¼ inch in inside diameter) attached. Measure 300 mL of water into the funnel. Open the funnel and allow the water to flow on the magnesia and screen while the magnesia is being brushed with the ½ inch brush. Regulate the flow of water so that the step above will take 2.0±0.1 minutes and so that as much of the sample as possible is brushed through the screen. Wash the magnesia from the brush on to the screen with the last 40 mL of water. Rinse the screen and residue with acetone and dry in an oven at 110˚C. Carefully brush out the residue and weigh on an analytical balance. Calculate the granulation as follows:

$$\%\ \text{retained on screen} = \frac{\text{weight of residue}}{\text{sample weight}}\ \text{x}\ 100$$

$$\%\ \text{passing through screen} = 100.00 - \%\ \text{retained on screen}$$

4.3.9 Aluminum, as aluminum oxide (Al_2O_3). Determine aluminum content in accordance with the following.

4.3.9.1 Apparatus and reagents.
a. Analytical balance
b. Hot plate
c. Muffle furnace (1600˚C)
d. Laboratory oven (110˚C)
e. Desiccator
f. Hydrochloric acid, reagent grade
g. Ammonium hydroxide, reagent grade
h. Methyl red indicator solution
i. Filter paper, Whatman No. 40
j. Porcelain crucibles, Coors No. 1, tall form
k. Ammonium nitrate, reagent grade, 1% solution

4.3.9.2 Procedure. Weigh approximately two grams of the material to the nearest 0.0001 g. Transfer the material to a 250-mL beaker. Add 5 mL distilled water and 20 mL concentrated hydrochloric acid with caution. Digest on a hot plate until solution is complete. Break up any lumps with a stirring rod and cover the beaker with a watch glass. Evaporate the solution to dryness. Cool the sample in air and add 5 mL concentrated hydrochloric acid. Warm for a few minutes on a hot plate, then add 50 mL of hot acidified water (5 mL of 1:1 HCl in 1 liter of distilled water). Filter the solution through No. 30 paper, into a 400-mL beaker, washing the residue with hot water until the filtrate reaches a volume of about 250 mL. To the filtrate add 5 g ammonium chloride, 3 drops of methyl red indicator and heat to boiling. Add concentrated ammonium hydroxide slowly, finally drop by drop until just alkaline (yellow to methyl red). Add a pinch of ashless filter pulp and boil for 3 minutes. Let stand for 15 minutes to coagulate

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the hydrous oxides. Filter the solution through No. 30 paper. Wash the residue 4-5 times with hot 1% ammonium nitrate solution. Transfer the filter paper to a previously ignited and tared crucible, dry in an oven (100˚C) for one hour. Char the filter paper with a low heat and finally ignite at red heat until no carbon remains. Ignite the crucible in the muffle for one hour. Cool in a desiccator and reweigh the crucible. Difference in weight is the precipitate. Calculate the iron oxide and aluminum oxide concentrations as follows:

$$R_2O_3 = \% Fe_2O_3 + Al_2O_3$$

$$\% R_2O_3 = \frac{\text{weight of precipitate} - \text{ash of paper}}{\text{sample weight}} \text{ x } 100$$

Using the iron concentration as determined in 4.3.2 Iron as Fe_2O_3, calculate the Al_2O_3 concentration as follows:

$$\% Al_2O_3 = \% R_2O_3 + Fe_2O_3$$

4.3.10 Moisture (Types II and III). Weigh approximately 20 g accurate to 0.1 mg of sample into a wide mouth, glass stoppered, low form drying dish (previously tared). Transfer to an oven at 105˚C. Dry for one hour. Remove from oven. Cool in a desiccator and reweigh. Calculate percent moisture as follows:

$$\% \text{ moisture} = \frac{A - B}{A} \text{ x } 100$$

Where: A = original sample weight, g
B = dried sample weight, g

4.3.11 Surface area (BET) (Type III). Verify surface area in accordance with ISO 9277.

4.3.12 Workmanship. Visually examine the contents of the container sampled and the sample taken for the presence of foreign materials.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The magnesium oxide covered in this use is intended for use in aerial mines and in the M864 and XM1128 propellants.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. Whether a first article sample is required (see 3.3).
d. Packaging requirements (see 5.1).

6.2.1 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, or as directed in the contract should be supplied by the contractor.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs, as required, to the Technical Agency (see 6.6)

6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer to the Technical Agency (see 6.6). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5 Approval of equivalent quality assurance systems. Prior approval of the Contracting Officer is required for use of equivalent quality assurance systems. A description of the proposed system should be submitted through the contracting officer to the Technical Agency (see 6.6). The contractor should submit a written description of the quality assurance system for approval prior to initiation of production. It should include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, and provisions for control and disposition of nonconforming material.

6.6 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, CCDC – AC, ATTN: FCDD-ACE-QMG, Picatinny Arsenal, NJ 07806-5000.

6.7 Subject term (key word) listing.

Igniter
Base bleed
Propellant

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Burn rate modifier

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army – AR

Preparing activity:
Army – AR
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